

Solution

WORK-03

Class 09 - Science

Section A

1.
(c) A is true but R is false.
Explanation: The quality or timbre of a sound is that characteristic which enables us to distinguish one sound from another having the same pitch and loudness. The sound which is more pleasant is said to be of rich quality.
2.
(c) A is true but R is false.
Explanation: The velocity of sound in a gas is directly proportional to the square root of its absolute temperature. Since the temperature of a hot day is more than a cold winter day, therefore sound would travel faster on a hot summer day than on a cold winter day.
3.
(b) Both A and R are true but R is not the correct explanation of A.
Explanation: The amplitude of a wave is the same as the amplitude of the vibrating body producing the wave. The loudness or softness of a sound is determined basically by its amplitude which depends upon the force with which an object is made to vibrate.
4.
(d) A is false but R is true.
Explanation: The velocity of sound in the gaseous medium is given by $v = \sqrt{\frac{\gamma P}{\rho}}$. Clearly, the velocity of sound in a gas is inversely proportional to the square root of the density of the gas. The density of oxygen is 16 times the density of hydrogen, therefore, the velocity of sound in hydrogen is four times the velocity of sound in oxygen.
5.
(b) Both A and R are true but R is not the correct explanation of A.
Explanation: When the speed of any object exceeds the speed of sound it is said to be traveling at supersonic speed. Bullets, jet aircraft, etc. often travel at supersonic speeds.
6.
(a) Both A and R are true and R is the correct explanation of A.
Explanation: In longitudinal wave, pressure variations travel along the tube with the density variations, when the density reaches its maximum at a particular location, the pressure also reaches its maximum at that location.
7.
(c) A is true but R is false.
Explanation: The sound energy gets partially absorbed by objects with soft and loose textures.
8.
(c) A is true but R is false.
Explanation: The propeller of a motorboat cuts the water surface laterally and also pushes it in a backward direction. Hence, it will result in both longitudinal and transverse waves.
9.
(b) Both A and R are true but R is not the correct explanation of A.
Explanation: The speed of sound is 340 m/s and speed of light is 3×10^8 m/s in air. Thus flash of lightning is seen before the sound of thunder is heard.
10.
(a) Both A and R are true and R is the correct explanation of A.
Explanation: Pitch is the physiological sensation that helps in distinguishing a shrill sound from a flat sound. The faster the vibration of the source, the higher is the frequency and the higher is the pitch. Thus, a high pitch sound corresponds to more number of compressions and rarefactions passing a fixed point per unit time.
11.
(c) A is true but R is false.

Explanation: If the tuning fork is held in hand and hit with a hammer, a sound is produced as the prongs of the tuning fork set surrounding air particles into vibrational motion. The sound produced by the tuning fork in the air is barely audible. However, if the tuning fork is set on a desktop. The desktop begins vibrating at the same natural frequency as the tuning fork. The tuning fork forces surrounding desktop particles into vibrational motion. The vibrating desktop in turn forces surrounding air particles into vibrational motion. As the surface area of the desktop is greater than the surface area of the tuning fork, more surrounding air particles will be forced into vibration. This causes an increase in the amplitude and thus loudness of the sound.

12. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Compression is a region of the medium in which particles are compressed i.e., particles come closer i.e., the distance between the particle becomes less than the normal distance between them. Thus there is a temporary decrease in volume and a consequent increase in density of the medium. Similarly, in rarefaction, particles get farther apart and a consequent decrease in density.

- 13.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: As we move up the pressure of air, density of air, both decrease. But, velocity of sound will not change as long as the temperature of air remains constant. Since temperature falls as we ascend up, hence the velocity of sound also decreases.

- 14.

- (d) A is false but R is true.

Explanation: Since mechanical transverse wave can propagate through medium which possesses elasticity of shape. Air possesses only volume elasticity therefore transverse wave cannot propagate through air.

- 15.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: The speed of sound tells us the rate at which the sound travels from the sound-producing body to our ears. The speed of sound depends on a number of factors. These are given below:

- i. The speed of sound depends on the nature of the material through which it travels.
- ii. The speed of sound depends on the temperature.
- iii. The speed of sound depends on the humidity of the air.

- 16.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: When the piston is moved back and forth, the molecules of the liquid also move along the direction of motion of the piston (direction of disturbance) also the volume and density of the liquid is changing with the motion of the piston.

17. (a) Both A and R are true and R is the correct explanation of A.

Explanation: If we stand in one corner of a big empty hall and shout the word 'hello' we will hear the word 'hello' coming from the empty hall in the form of an echo a little while later. It appears as if the hall is repeating our 'hello'. This happens because the sound of our 'hello' is reflected from the walls of the hall and this reflected sound forms the echo.

- 18.

- (c) A is true but R is false.

Explanation: Velocity of sound increases with increase in humidity so it depends on the medium.

19. (a) Both A and R are true and R is the correct explanation of A.

Explanation: The velocity of sound in solid is given by, $v = \sqrt{\frac{E}{\rho}}$ Though ρ is large for solids, but their coefficient of elasticity E is much larger (compared to that of liquids and gases). That is why v is maximum in case of solid.

20. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Sound waves required a material medium to travel. As there is no atmosphere, i.e., vacuum is on the surface of the moon, therefore the sound waves cannot reach from one person to another. Because of this reason, for communication on the moon astronomers use phones, which convert sound waves into radio waves (electromagnetic waves need no medium for propagation).

21. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Sound needs a material medium like solid, liquid, or a gas to travel and be heard. Sound waves are mechanical waves because it require a material medium for their propagation.

22. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: When a vibrating object moves forward, it pushes and compresses the air in front of it creating a region of high pressure. As the object moves back and forth rapidly, a series of compressions and rarefactions are created in the air. These make the sound wave that propagates through the medium.
 The energy required to make an object vibrate is provided by some outside source like our hand, wind, etc.

23. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.

24. (a) Both A and R are true and R is the correct explanation of A.
Explanation: A sound is a form of energy that makes us hear. It travels in the form of waves. The disturbance which moves through a medium when the particles of the medium set the neighbouring particles into motion is known as a wave.

Section B

25. (d) C
Explanation: Speed = $\frac{\text{Distance}}{\text{Time}}$
 Here
 distance = 8m
 Time taken is given as: $t = \frac{10}{6}$ secs
 Speed = $\frac{8}{\frac{10}{6}} = 8 \times \frac{6}{10} = 4.8 \text{ m/s}$

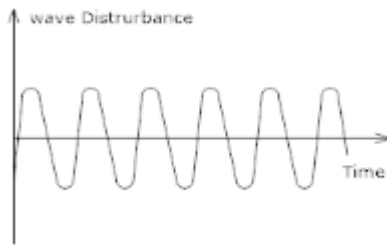
Section C

26. (c) 2 : 1
Explanation: In case of coherent sources, interference occurs. Intensity at the midpoint is
 $I_{\max} = k(a + a)^2 = k \times 4a^2$
 In case of incoherent sources, interference does not occur.
 Intensity at the midpoint is
 $I = I_1 + I_2 = ka^2 + ka^2 = k \times 2a^2$
 $\therefore \frac{I_{\max}}{I} = \frac{4}{2} = 2 : 1$

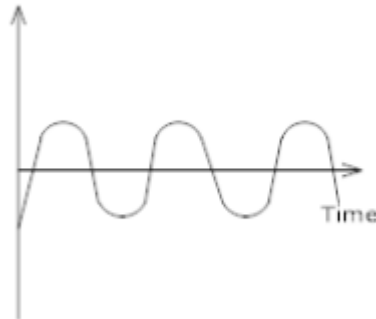
Section D

27. In this situation, longitudinal waves will produce in spring.
 28. 20 Hz to 20,000Hz.
 29. Sound is a form of energy that produces a sensation of hearing in our ears.
 Sound is produced when any object vibrates/oscillates.
 30. Waves that are characterised by the motion of particles in a medium are called mechanical waves. Mechanical waves require material medium for their propagation.
 31. A wave is a vibratory disturbance in a medium which carries energy from one point to another without being in direct contact between the two points.
 32. Sound wave is called a longitudinal wave because sound waves travel in the air through compressions and rarefactions.
 33. Sound will travel the fastest in iron at a particular temperature because the particles in solid are closely packed and the sound wave jumps off each particle fastly.
 34. According to law of reflection $\angle i = \angle r$
 Since $\angle i = 40^\circ$, θ the angle of reflection should be 40°
 35. Speed of sound = distance/time
 therefore, distance travelled by sound during echo = $\text{speed} \times \text{time} = 342 \times 3 = 1026 \text{ m}$
 so the distance of reflecting surface = $\frac{1026}{2} = 513 \text{ m}$
 36. Infrasonic waves are produced in an earthquake before the main shock wave begins.

37. High pitch sound is those sound which has a higher frequency that is in 1 second they complete a large number of vibrations.



Low pitch sound is those sound which has a lesser frequency that is in 1 second they complete less number of vibrations.



38. a. **Infrasound** : Sound waves between the frequencies 1 to 20 Hz.

b. **Ultrasound**: Sound waves of the frequencies above 20,000 Hz.

39. Pitch of the sound wave.

40. Frequency = 220 Hz

$$\text{Speed of sound} = 440 \text{ ms}^{-1}$$

We know speed of sound = Frequency $\times$ wavelength

$$= 220 \times \text{wavelength}$$

$$\Rightarrow \text{Wavelength} = \frac{440}{220} = 2\text{m}$$

41. The amount of sound energy passing through unit area each second is called the intensity of sound.

42. Graph represented by 'a' is of male voice. We know that male voice has lower frequency than female voice, and graph shown by 'a' shows lower frequency.

43. The sound wave will travel faster in steel because speed of sound is the fastest in the solids.

44. We have given that,

$$\text{Speed of sound, } v = 346 \text{ m s}^{-1}$$

$$\text{Time taken for hearing the echo, } t = 2\text{s}$$

Therefore, Distance travelled by the sound

$$= v \times t = 346 \text{ ms}^{-1} \times 2 \text{ s} = 692 \text{ m}$$

In 2 s sound has to travel twice the distance between the cliff and the person. Hence, the distance between the cliff and the person

$$\frac{692\text{m}}{2} = 346 \text{ m}$$

45. Speed of sound = 340 m/s and time = 10 s

$$\text{Distance} = \text{speed} \times \text{time} = 340 \times 10 = 3400 \text{ m}$$

46. There is no atmosphere on moon and we know that sound waves need a medium to travel. Thus, we do not hear the sound of an explosion on the surface of the Moon.

47. The change in density from one maximum value to the minimum value and again to the maximum value makes one complete oscillation.

48. To reduce reverberation, the roofs and walls of an auditorium or hall are generally covered with sound absorbent materials.

49. Longitudinal waves; because sound waves are longitudinal waves.

50. Sound of Guitar has a higher pitch.

51. Given:

$$\text{Distance between the crest and next trough} = 10$$

$$\text{or } \lambda/2 = 10$$

$$\lambda = 20\text{cm} = 0.20\text{m}$$

$$\text{velocity } v = u \times \lambda$$

$$= 12 \times 0.20$$

$$= 2.40 \text{ m/s}$$

52. a. To hear the sound of the coming train because sound travel within the rail track and get the idea of distance of incoming train.
b. Copper.
c. For safety as he cared for his friend.
53. a. The technique of obtaining picture of internal organs of body by using echoes of ultrasound pulse is called ultrasonography.
b. The ultrasonography is based on the principle that ultrasound waves are sent from transducer and propagate through different tissues and then return to the transducer as reflected echoes.
c. To disclose the gender is against the law and also to discourage the curiosity of knowing the sex of the child before birth.
d. Obeying laws, honesty.
54. Since speed of thunder (sound) is much less (332 m/s) as compared to speed of flash (light) which is about 3×10^8 m/s therefore light travels faster than sound hence thunder is heard a few seconds after the flash is seen.
55. i. In the graphical figure, both the waves are of the same frequency but with different amplitude. In wave, A amplitude is less and in wave, B amplitude is higher.
ii. The relationship between the velocity of sound, its wavelength and frequency is given as-
Velocity of sound = frequency $\times$ wavelength
iii. Amplitude
iv. Frequency its unit is hertz.
56. i. (d)
ii. Intellectual honesty, desire to know more and improve.
iii. Submitting honest information for income tax returns, honest dealings.
57. i. The two graphical figure represents the same amplitude but different frequency.
ii. The pitch of the sound is determined by the frequency wave.
iii. Lower pitch indicates low frequency and higher pitch indicates the higher frequency that is $Pitch \propto Frequency$.
58. i. Ultrasound is used to monitor the development of the foetus inside the mother's womb.
ii. The ultrasound scanner transmits ultrasound into the mother's body and receives echoes formed by the reflection of ultrasound from inside. The ultrasound echoes form a picture of a developing baby on a monitor which helps the doctor to keep a track of the developing baby. Thus ultrasonography is used for the examination of the foetus during pregnancy to detect any growth and abnormalities.
iii. The sound above 20,000Hz is regarded as ultrasound
59. Since speed of sound in air = 344 m/s
and speed of sound in aluminium = 6420 m/s
we know that $v = \text{distance}/\text{time}$ therefore $\text{time} = d/v$
time taken by sound wave in air/time taken by sound wave in aluminium
 $= \frac{d}{344} : \frac{d}{6420} = \frac{6420}{344} = \frac{18.66}{1}$
the sound will take 18.66 times more time through air than in aluminium in reaching other boy.

60. i. The difference between the longitudinal wave and the transverse wave is as follows:

Longitudinal wave	Transverse wave
It needs medium for propagation	It may or may not needs a medium for propagation.
Particles of the medium move in a direction parallel to the direction of the propagation of disturbance ex-sound wave	Particles of the medium move in the perpendicular direction of the propagation of disturbance. Ex-light wave

- ii. The three characteristics of sound waves are
a. loudness
b. pitch
c. quality/timbre
- iii. The peak is called the crest and valley is called the trough in the given diagram.
61. i. Sound is produced when a material body is made to vibrate with some mechanical energy. So, school bell is struck with a hammer to make it vibrate and thus the sound is produced.
ii. The time taken by the signal to reach the bottom and come back is 4.8 s
Speed of sound in water = 1500 m/s
Depth of the ocean h
 $h = \frac{v \times t}{2}$

$$h = \frac{1500 \times 4.8}{2}$$

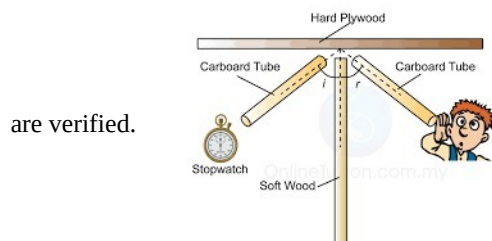
$$h = 1500 \times 2.4$$

$$h = 3600 \text{ m}$$

62. 1) The phenomenon of bouncing back of a sound wave from a surface (i.e. going from one medium to the same medium after striking the second medium) is known as reflection of sound.

2) Activity:

- Take two identical pipes, as shown in figure.
- The length of the pipes should be sufficiently long.
- Arrange them on a table near a wall.
- Put a clock near the open end of the one pipe as shown in figure.
- Now, try to hear the sound through the other pipe.
- Adjust the position of the pipe to hear best sound of clock.
- Now, measure the angles of incidence and reflection and see the relationship between the angles.
- The angle of incidence and the angle of reflection of sound are found to be equal.
- Also, Tubes are placed normal to the vertical reflector, and the reflected wave lie on the same plane. Hence law of reflection



3) Laws of reflection:

- The angle of incidence of sound wave is always equal to the angle of reflection of sound wave.
- The incident sound wave, reflected sound wave and the normal to the reflecting surface at the point of incidence, all lie in the same plane.

63. Frequency: The number of complete sound waves (or oscillations) produced in one second is called frequency of the sound wave.

Wavelength: The minimum distance in which a sound wave repeats itself.

Echo: The repetition of sound caused by the reflection of sound waves is called an echo.

The full form of SONAR is Sound Navigation and Ranging.

Two applications of ultrasound are:

- Ultrasound is used in industries for detecting flaws in metal blocks without damaging them.
- Ultrasound is used to investigate the internal organs of the human body such as liver, gall bladder, pancreas, kidneys, etc.

64. i. The number of waves produced per second is called the frequency of the wave.

ii. Sound waves range from 20 Hz to 20 kHz

iii. a. Since the distance between a crest and next trough is $\frac{\lambda}{2}$.

Therefore, $\frac{\lambda}{2} = 50 \text{ cm}$ (given)

$\lambda = 100 \text{ cm}$ or 1 m

b. One vibration or oscillation consists of one trough and one crest.

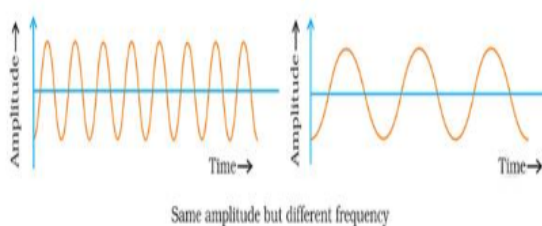
In 0.2s number of vibrations are 20

So, in 1s number of vibrations = $\frac{20}{0.2} = 100$

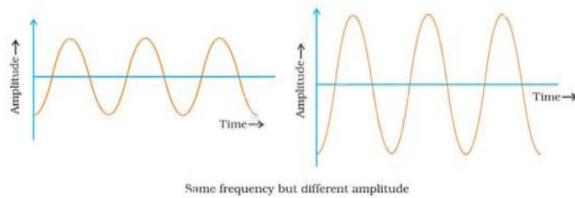
$\therefore$ Frequency = 100 Hz

c. Time period, $T = \frac{1}{\text{frequency}} = \frac{1}{100} = 0.01 \text{ sec}$

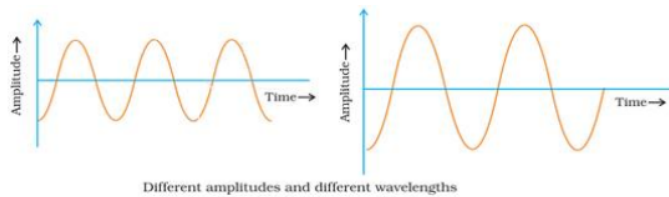
65. i.



ii.

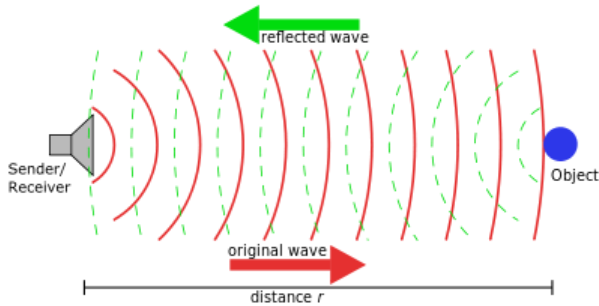


iii.



66. i. Timbre or quality of sound that is distinct from its pitch and intensity helps to identify your friend by his voice while sitting with others in a dark room.
- ii. Frequency = $\frac{1}{\text{Time Period}}$
 $\lambda = 4 \text{ cm} = 0.02 \text{ m}$, $v = 20 \text{ m/s}$, $V = ?$ and $T = ?$
 $V = \frac{v}{\lambda} = \frac{20}{0.04} = 500 \text{ Hz}$
 $T = \frac{1}{V} = \frac{1}{500} = 0.002 \text{ s}$
67. i. The full name of SONAR is Sound Navigation and Ranging. Sonar is based on the principle of reflection of sound wave. Powerful pulses of ultrasound are sent out at regular intervals from a transmitter mounted on a ship. When these pulses are intercepted by submerged objects, they get reflected. The reflected sound or echo is detected by an underwater receiver, which is also mounted on the ship. If speed of ultrasound be v and t is the elapsed time between the transmission and the reception of the ultrasound signal, the depth of the submerged object underwater is $h = \frac{v \times t}{2}$
- ii. $t = 5 \text{ s}$, $h = 2625 \text{ m}$, $v = ?$
 $h = \frac{v \times t}{2}$
 $2625 = \frac{v \times 5}{2}$
 $v = \frac{2625 \times 2}{5}$
 $v = 1050 \text{ m/s}$
68. i. The speed of sound tells us the rate at which the sound travels from the sound-producing body to our ears. The speed of sound depends on a number of factors. These are given below:
- The speed of sound depends on the nature of the material through which it travels.
 - The speed of sound depends on the temperature.
 - The speed of sound depends on the humidity of the air.
- ii. Bats search their prey at night by the method of echolocation. They emit high-frequency ultrasonic squeaks while flying and listen to the echoes produced by the reflection of their squeaks from the prey like a flying object. From the time taken by the echo to be heard, bats can judge the distance of the insect and hence catch it.
69. Sonar is a device that uses ultrasonic waves to measure the distance, direction, and speed of underwater objects. Sonar consists of a transmitter and a detector and is installed in a boat or a ship. The transmitter produces and transmits ultrasonic waves. These waves travel through water and after striking the object on the seabed, get reflected back and are sensed by the detector. The detector converts the ultrasonic waves into electrical signals which are appropriately interpreted. The distance of the object that reflected the sound wave can be calculated by knowing the speed of sound in water and the time interval between transmission and reception of the ultrasound. Let the time interval between transmission and reception of ultrasound signal be t and the speed of sound through seawater be v . The total distance, $2d$ travelled by the ultrasound is then, $2d = v \times t$. The above method is called echo-ranging. The sonar technique is used to determine the depth of the sea and to locate underwater

hills, valleys, submarine, icebergs, sunken ship etc.



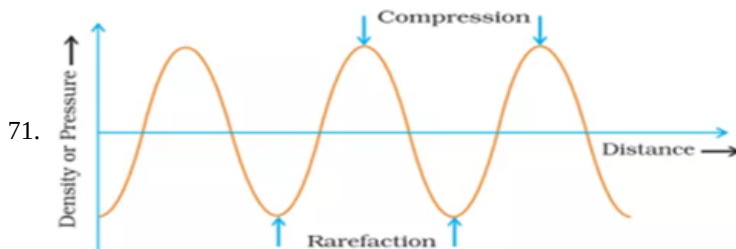
70. i. Echo ranging is the process of detecting the underwater objects/obstacles by using the sound signals. For the formation of the echo, the minimum distance between the source and the reflecting body should be 17 metres.
Echo ranging technique is used to measure the depth of the sea with the help of the SONAR (Sound navigation and ranging) system.

ii. Given: Wavelength, $\lambda = 20 \text{ cm} = 0.20 \text{ m}$ and wave velocity (v) = 36 m/s

a. We know that $v = n\lambda$

$$\Rightarrow n = \frac{v}{\lambda} = \frac{36}{0.20} = 180 \text{ waves per second.}$$

b. Time period $T = \frac{1}{v} = \frac{1}{180} = 5.55 \times 10^{-3} \text{ seconds.}$



Wavelength is the distance between two consecutive compressions or two consecutive rarefactions of a wave sound. Wavelength is represented by the Greek letter λ (lambda). The SI unit of wavelength is metre (m).

Time period is the time taken to travel the distance between any two consecutive compression or rarefaction from a fixed point.

The time period of a sound wave is represented by letter 'T'. The SI unit of time period is second (s).

72. i. The repetition of sound caused by the reflection of sound waves from an obstacle is known as an echo whereas the phenomenon of persistence or prolongation of audible sound after the source has stopped emitting it is called reverberation.
ii. In an auditorium or big hall, excessive reverberation is highly undesirable. To reduce reverberation, the roof and walls of the auditorium are generally covered with sound-absorbent materials like compressed fibreboard, rough plaster or draperies.
iii. The soft and the porous materials are bad reflectors of sound but are good absorbers of sound.
iv. We can hear more clearly in a room having curtains because curtains are bad reflectors of sound. The curtains absorb most of the sound falling on them, and hence do not produce echoes.
73. i. A megaphone works on the principle of multiple reflections of sound. When a person speaks into the narrow end of the megaphone tube, the sound waves produced by his voice are prevented from spreading by successive reflections from the wider end of the megaphone tube. Due to this, the sound of the person can be heard over a longer distance.

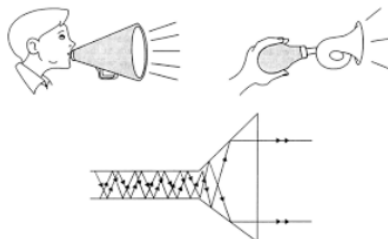


Figure- Multiple reflection sound in the megaphone.

- ii. The stethoscope is a medical instrument used by doctors for listening to the sounds produced within the human body, mainly in the heart and the lungs. The doctor puts the earpiece of the stethoscope into his ears and places the chest-piece above the part of the patient's body which is to be examined. The sound of heartbeats reaches the doctor's ears by the multiple reflections of the sound waves through the stethoscope tube. Thus, it works on the principle of multiple reflections of sound.

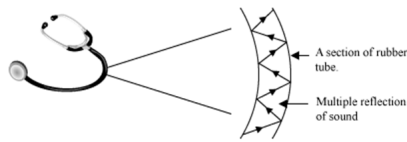
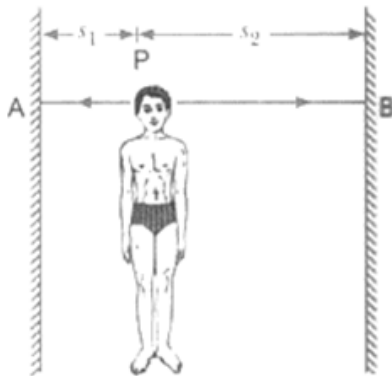


Figure- Multiple reflection sound in the stethoscope.

74. i. Let P be the person standing between the cliffs A and B. Let s_1 be a distance of nearest cliff A from P and s_2 the distance of second cliff B from P. The first echo is heard when sound reaches the person after being reflected from cliff A.



Given, $s_1 = AP = 640$ m

Time interval of first echo, $t_1 = 4$ seconds

From relation, $2s_1 = v t_1$, we have

$$\text{The speed of sound, } v = \frac{2s_1}{t_1}$$

$$\frac{2 \times 640}{4}$$

Therefore, Speed of sound in air, $v = 320$ m/s

- ii. The second echo is heard when the sound reaches the person after being reflected from the cliff B.

Time interval of second echo, $t_2 = 4 + 3 = 7$ seconds

Therefore, From relation, $2s_2 = v t_2$,

$$\text{We have, } \frac{vt_2}{2}$$

$$\frac{320 \times 7}{2}$$

$$= 1120 \text{ m}$$

Therefore, Distance between cliffs A and B,

$$s = s_1 + s_2 = 640 + 1120 = 1760 \text{ m}$$